"Problem Set 2 by Jimmy Zheng"

Question 1: ISL Chapter 2: Exercises 2,3,7.

Exercise 2)

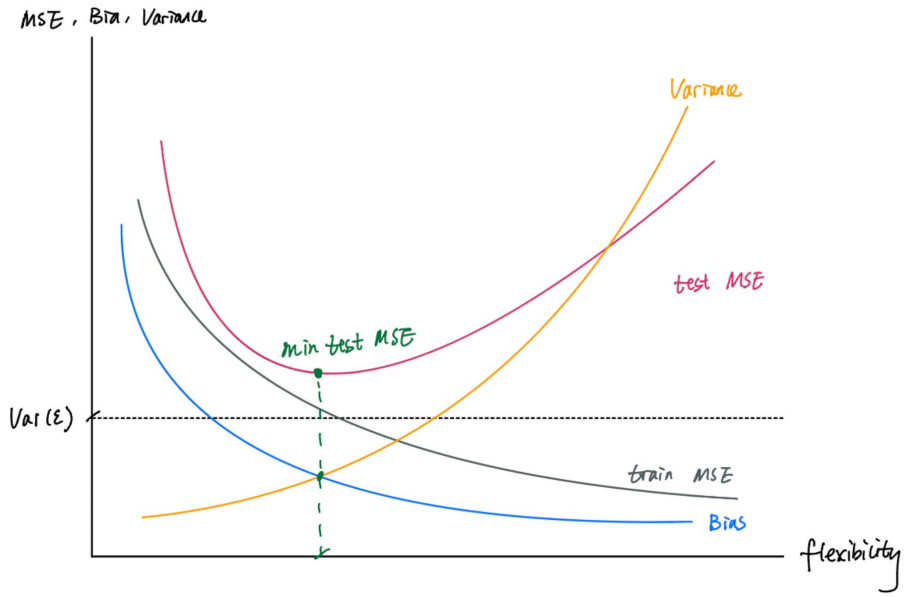
(a) This is a regression problem as the response variable, the CEO's salary, is quantitative. n=500 and p=3, which are the profit, number of employees, and industry. We are more interested in inference as we want to test hypotheses, such as whether the firm's profit significantly affects the CEO's salary, rather than predict the salary level.

(b) This is a classification problem as the response variable, success or failure, is qualitative. n=20 and p=13, which are the price, marketing budget, competition price, and ten other variables. We are more interested in prediction as we want to predict the success or failure rate to estimate the potential risks of launching the new product.

(c) This is a regression problem as the response variable, the % change in USD/Euro, is quantitative. n=52 and p=3, which are the % change in the US market, the % change in the British market, and the % change in the German market. We are more interested in prediction as we want to precisely forecast the change in the exchange rate to make financial decisions, which is crucial to our capital gains from the foreign exchange market.

Exercise 3)

(a)



(b) First of all, as flexibility increases, the model matches and fits the in-sample data more closely, striving to capture all patterns or the errors, so the training MSE always declines monotonically. However, The relative rate of change of bias and variance determines whether the test MSE

increases or decreases. Variance increases with higher flexibility as the highly flexible model tends to track the training data points overly closely, so changing any one of the sample data points may result in completely different estimated models. Nevertheless, as we increase the flexibility, we usually arrive at more accurate estimates of the true model, so the bias between the estimated and true model tends to initially decrease faster than the variance increases. Consequently, the expected test MSE declines at first. For example, using a thin-plate spline instead of linear regression model to fit the data significantly decreases the bias at first. However, at some point increasing flexibility, such as the smoothness of the spline, has little impact on the bias but starts to significantly increase the variance. When this happens the test MSE increases. However, adjusting the flexibility of the model could only at best make the reducible error of the model close to zero, the irreducible error, Var(ε), always exists and cannot be predicted by the predictors in the model. As a result, test MSE will never be lower than Var(ε).

Exercise 7)

(a)

For obs 1, Red, D1 = $\sqrt{(0-0)^2+(3-0)^2+(0-0)^2} = 3$

For obs 2, Red, D2 = $\sqrt{(2-0)^2+(0-0)^2+(0-0)^2} = 2$

For obs 3, Red, D3 = $\sqrt{(0-0)^2+(1-0)^2+(3-0)^2} = \sqrt{10}$

For obs 4, Green, D4 = $\sqrt{(0-0)^2+(1-0)^2+(2-0)^2} = \sqrt{5}$

For obs 5, Green, D5 = $\sqrt{(-1-0)^2+(0-0)^2+(1-0)^2} = \sqrt{2}$

For obs 6, Red, D6 = $\sqrt{(1-0)^2+(1-0)^2+(1-0)^2} = \sqrt{3}$

(b) When K=1, the prediction is green as the nearest point is observation 5 with a distance of $\sqrt{2}$ only.

(c) When K=3, the prediction is red. Among the nearest 3 points (observation 5, 6, and 2), 2/3 of them is red, and 1/3 of them is green. Thus, we predict it as red to maximize the conditional probability when X1=X2=X3=0.

(d) For highly non-linear boundaries, we expect the best value for K tends to be small. Smaller K means greater flexibility, which usually corresponds to lower bias, so the estimated decision boundary is more likely to be non-linear and closer to the true one. As K grows, the method becomes less flexible and produces a decision boundary that is close to linear. This corresponds a low-variance but high-bias classifier.

Problem Set 2 (Question 2 and 3) by Jimmy Zheng

Question 2: ISL Chapter 2 lab Section 2.3.4 (R loading of Auto.data)

```
library(ISLR2)
auto = read.table("Auto.data")
View(auto)
head(auto)
```

```
##     V1        V2           V3         V4     V5           V6   V7     V8
## 1  mpg cylinders displacement horsepower weight acceleration year origin
## 2 18.0         8        307.0      130.0  3504.         12.0   70      1
## 3 15.0         8        350.0      165.0  3693.         11.5   70      1
## 4 18.0         8        318.0      150.0  3436.         11.0   70      1
## 5 16.0         8        304.0      150.0  3433.         12.0   70      1
## 6 17.0         8        302.0      140.0  3449.         10.5   70      1
##                           V9
## 1                       name
## 2 chevrolet chevelle malibu
## 3         buick skylark 320
## 4        plymouth satellite
## 5             amc rebel sst
## 6               ford torino
```

```
auto = read.table("Auto.data", header = T, na.strings = "?", stringsAsFactors = T)
```

```
auto = read.csv("Auto.csv", na.strings = "?", stringsAsFactors = T)
View(auto)
head(auto)
```

```
##   mpg cylinders displacement horsepower weight acceleration year origin
## 1  18         8          307        130   3504         12.0   70      1
## 2  15         8          350        165   3693         11.5   70      1
## 3  18         8          318        150   3436         11.0   70      1
## 4  16         8          304        150   3433         12.0   70      1
## 5  17         8          302        140   3449         10.5   70      1
## 6  15         8          429        198   4341         10.0   70      1
##                       name
## 1 chevrolet chevelle malibu
## 2         buick skylark 320
## 3        plymouth satellite
## 4             amc rebel sst
## 5               ford torino
## 6          ford galaxie 500
```

```r
# set variable "origin" to as a categorical variable
auto$origin = as.factor(auto$origin)
```

```r
dim(auto)
```

```
## [1] 397   9
```

```r
auto = na.omit(auto)
dim(auto)
```

```
## [1] 392   9
```

```r
names(auto)
```

```
## [1] "mpg"          "cylinders"    "displacement" "horsepower"   "weight"
## [6] "acceleration" "year"         "origin"       "name"
```

Question 3: ISL Chapter 2: Exercises 9 (Ranalysis of Auto.data, which is a continuation of the lab section)

(a) Which of the predictors are quantitative, and which are qualitative?

```r
sapply(auto, class)
```

```
##          mpg    cylinders displacement   horsepower       weight acceleration
##    "numeric"    "integer"    "numeric"    "integer"    "integer"    "numeric"
##         year       origin         name
##    "integer"     "factor"     "factor"
```

"origin" and "name" are qualitative variables. "mpg", "cylinders", "displacement", "horsepower", "weight", "acceleration" and "year" are quantitative variables.

(b) What is the range of each quantitative predictor? You can answer this using the range() function.

```r
sapply(auto[ , 1:7], range)
```

```
##       mpg cylinders displacement horsepower weight acceleration year
## [1,]  9.0         3           68         46   1613          8.0   70
## [2,] 46.6         8          455        230   5140         24.8   82
```

(c) What is the mean and standard deviation of each quantitative predictor?

```r
# mean
sapply(auto[ , 1:7], mean)
```

```
##          mpg    cylinders displacement   horsepower       weight acceleration
##    23.445918     5.471939   194.411990   104.469388  2977.584184    15.541327
##         year
##    75.979592
```

```r
# standard deviation
sapply(auto[ , 1:7], sd)
```

```
##          mpg    cylinders displacement   horsepower       weight acceleration
##     7.805007     1.705783   104.644004    38.491160   849.402560     2.758864
##         year
##     3.683737
```

(d) Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?

```r
auto_subset = auto[-c(10:85),]

# range
sapply(auto_subset[ , 1:7], range)
```

```
##       mpg cylinders displacement horsepower weight acceleration year
## [1,] 11.0         3           68         46   1649          8.5   70
## [2,] 46.6         8          455        230   4997         24.8   82
```

```r
# mean
sapply(auto_subset[ , 1:7], mean)
```

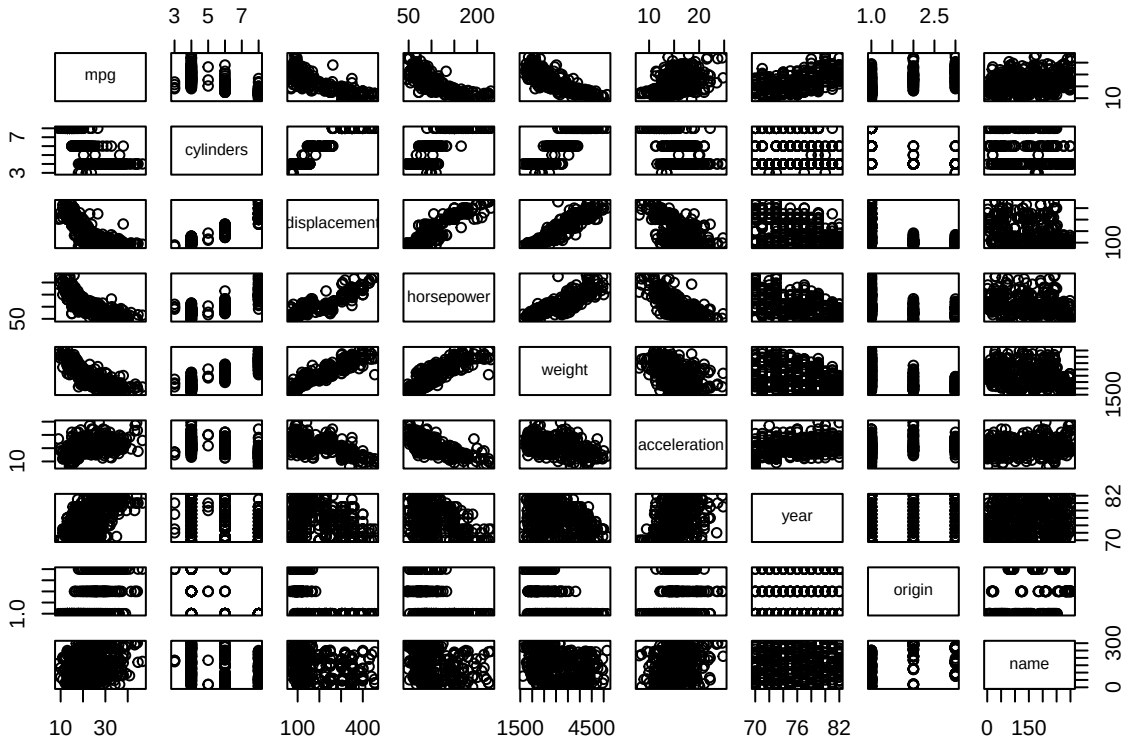
```
##          mpg    cylinders displacement   horsepower       weight acceleration
##    24.404430     5.373418   187.240506   100.721519  2935.971519    15.726899
##         year
##    77.145570
```

```r
# standard deviation
sapply(auto_subset[ , 1:7], sd)
```

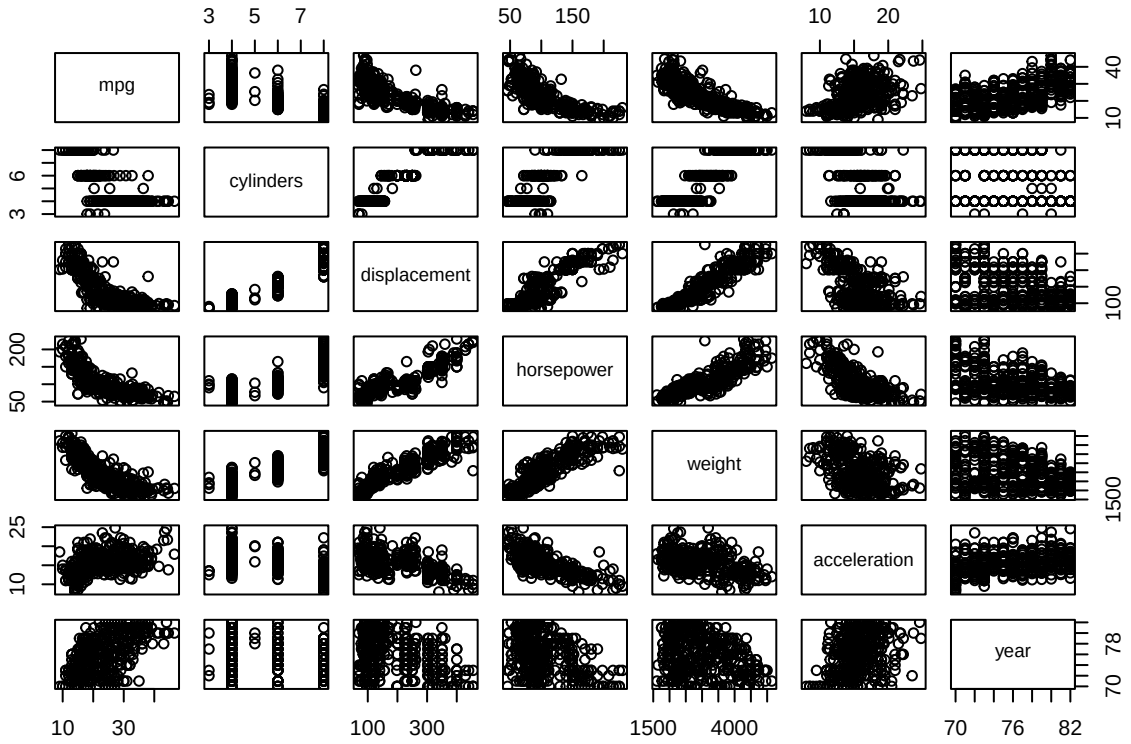
```
##          mpg    cylinders displacement   horsepower       weight acceleration
##     7.867283     1.654179    99.678367    35.708853   811.300208     2.693721
##         year
##     3.106217
```

(e) Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.

```r
pairs(auto)
```




```
# quantitative predictors only
pairs(auto[ , 1:7])
```



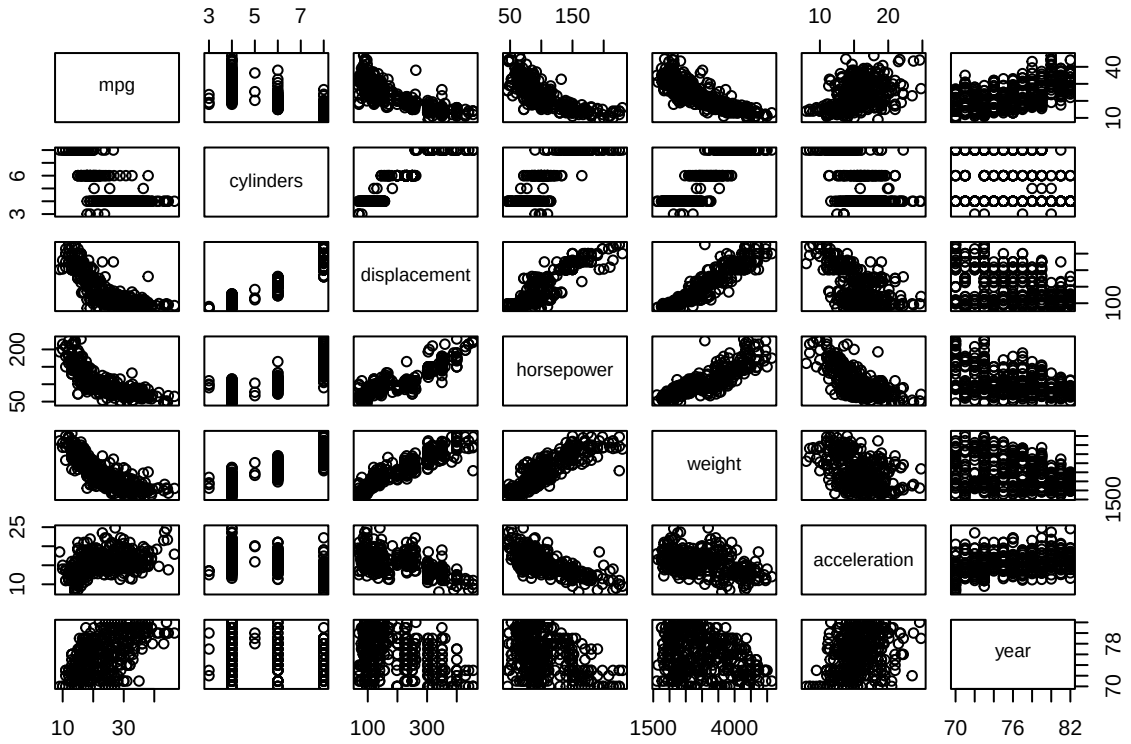
From the pairwise scatter plots, we observe that several factors related to fuel consumption, such as engine "displacements," "cylinders," "horsepower," and "weight," are generally positively correlated with each other. This makes sense as we generally assume that sophisticated and larger cars are usually heavier and equipped with more cylinders and engines with higher displacements and horsepower, resulting in higher fuel consumption.

As for their relationship with "mpg," we observe that "displacements," "cylinders," "horsepower," and "weight" all tend to have a negative relationship with it since higher fuel consumption may correspond to lower efficiency.

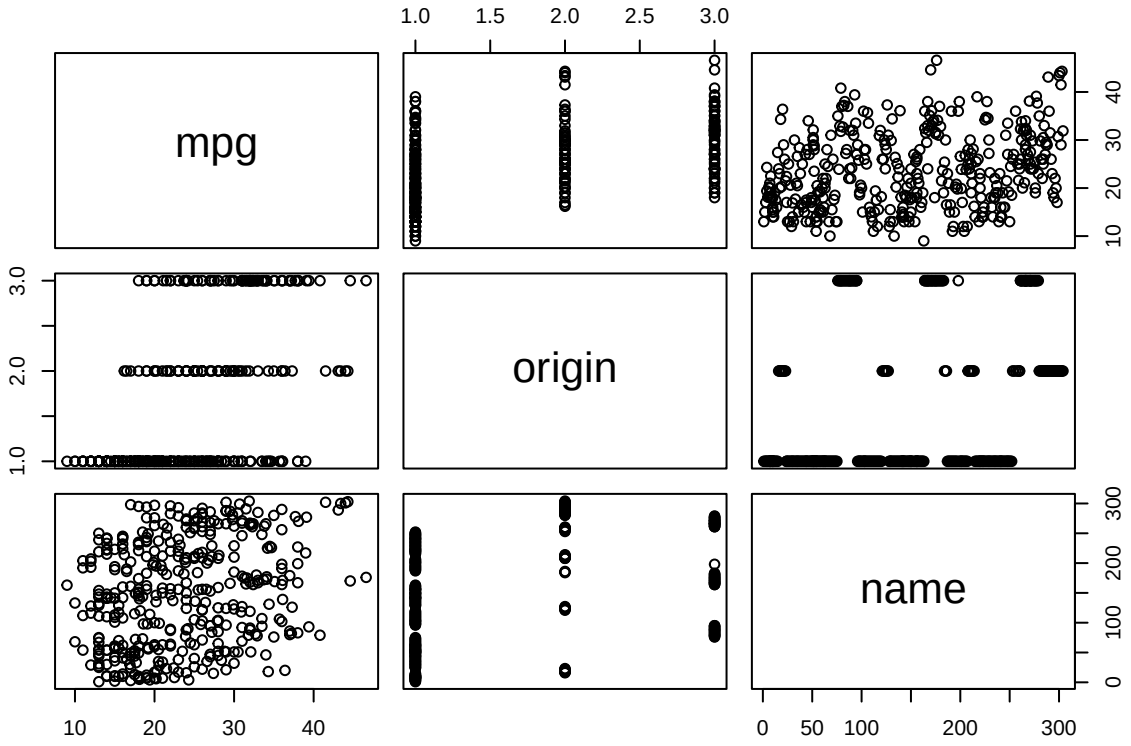
The relationship between qualitative variables ("origin" & "name") with other quantitative variables is less apparent.

(f) Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg? Justify your answer.

```
# quantitative predictors only
pairs(auto[ , 1:7])
```




```
# qualitative predictors only
pairs(auto[ , c(1,8:9)])
```




```
cor(auto[,1:7])
```

```
##                     mpg  cylinders displacement horsepower     weight
## mpg           1.0000000 -0.7776175   -0.8051269 -0.7784268 -0.8322442
## cylinders    -0.7776175  1.0000000    0.9508233  0.8429834  0.8975273
## displacement -0.8051269  0.9508233    1.0000000  0.8972570  0.9329944
## horsepower   -0.7784268  0.8429834    0.8972570  1.0000000  0.8645377
## weight       -0.8322442  0.8975273    0.9329944  0.8645377  1.0000000
## acceleration  0.4233285 -0.5046834   -0.5438005 -0.6891955 -0.4168392
## year          0.5805410 -0.3456474   -0.3698552 -0.4163615 -0.3091199
##              acceleration       year
## mpg             0.4233285  0.5805410
## cylinders      -0.5046834 -0.3456474
## displacement   -0.5438005 -0.3698552
## horsepower     -0.6891955 -0.4163615
## weight         -0.4168392 -0.3091199
## acceleration    1.0000000  0.2903161
## year            0.2903161  1.0000000
```

Among all the quantitative variables, “displacement,” “horsepower,” and “weight” seem to have the most apparent negative relationship with “mpg.” In fact, their sample correlations with “mpg” are indeed the highest, which are -0.8051269, -0.7784268, and -0.8322442 respectively. This is reasonable since heavier vehicles with more powerful engines tend to have greater inertia and energy consumption, contributing to decreased efficiency.

However, qualitative variables, such as “name,” are less favorable. In general, no apparent relationship between “mpg” and “name” could be observed from the scatter plot. Nevertheless, I would like to include the product of “origin” and any quantitative variable as interaction terms in the model to identify the synergy effects. From the scatter plot, Japanese cars (origin=3) seem to have higher “mpg” than American and European cars.